

# DJEZZY NATIONAL SOC PLATFORM

Operations Manual

Version 2.0 | Day-to-Day Operations Guide

Effective Date: 2026-07-30

Audience: SOC Analysts, Incident Responders, Security  
Engineers

**OPERATIONAL STATUS: READY FOR 24/7 OPERATIONS | ALL PROCEDURES VALID**

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# 1. Daily Operations Checklist

Every shift begins with a systematic review of platform status, pending items from previous shift, and awareness of any ongoing incidents or investigations. This checklist ensures consistent operational coverage and prevents tasks from falling through cracks during shift transitions. Complete all items within the first 30 minutes of shift start, documenting results in the shift log. Any item marked as FAIL requires immediate escalation to shift lead or on-call engineer before proceeding with other duties.

## 1.1 Shift Start Procedures

| #  | Task                             | Frequency    | Tool/Command              | Pass Criteria               |
|----|----------------------------------|--------------|---------------------------|-----------------------------|
| 1  | Review platform health dashboard | Each shift   | Grafana SOC Overview      | All green                   |
| 2  | Check overnight alerts summary   | Each shift   | Email/TheHive inbox       | Zero critical unacked       |
| 3  | Verify backup completion         | Daily        | Backup logs               | Last backup < 24h ago       |
| 4  | Review disk utilization          | Daily        | df -h / docker system df  | < 80% on all volumes        |
| 5  | Check SSL certificate expiry     | Weekly       | openssl x509 -checkend... | > 30 days remaining         |
| 6  | Review user access changes       | Each shift   | Auth logs / LDAP audit    | No unauthorized changes     |
| 7  | Verify DR replication lag        | Daily        | pg_stat_replication       | lag < 1 minute              |
| 8  | Review failed login attempts     | Each shift   | auth.log / lastb          | Investigate spikes > normal |
| 9  | Check Kafka consumer lag         | Each shift   | Kafka Exporter dashboard  | lag < 1000 messages         |
| 10 | Document shift handover notes    | End of shift | Shift log template        | Complete handover doc       |

## 1.2 Weekly Tasks

| Day       | Tasks                                            | Responsible    | Duration  |
|-----------|--------------------------------------------------|----------------|-----------|
| Monday    | Review weekend alerts, patch assessment review   | Shift Lead     | 2 hours   |
| Tuesday   | Rule tuning based on FP/NF analysis              | Senior Analyst | 3 hours   |
| Wednesday | Threat intel feed review, IOC updates            | Intel Analyst  | 2 hours   |
| Thursday  | Vulnerability scan review, remediation tracking  | Vuln Analyst   | 3 hours   |
| Friday    | Week summary, metrics report, planning next week | Team Lead      | 2 hours   |
| Saturday  | Maintenance window (rotating), system updates    | On-call Eng    | 4 hours   |
| Sunday    | Reduced staffing, emergency response only        | Weekend Ops    | As needed |

## 2. Alert Triage Procedures

Alert triage is the foundational activity of security operations, determining which of the thousands of daily alerts warrant investigation and response. Effective triage balances thoroughness with efficiency—missing a real attack is unacceptable, but spending equal time on every alert is unsustainable at scale. The Djezzy SOC Platform processes alerts from 15 integrated security tools, each with different severity scales and context requirements. This section establishes standardized triage procedures that ensure consistent, defensible decisions across all analysts regardless of experience level.

### 2.1 Alert Severity Classification

| Severity      | Definition                                         | Response Time | Examples                                                  | Escalation                   |
|---------------|----------------------------------------------------|---------------|-----------------------------------------------------------|------------------------------|
| Critical (P1) | Active breach, data exfiltration, ransomware       | < 15 minutes  | Domain controller compromise, payment system breach       | Immediate escalation to CISO |
| High (P2)     | Significant compromise indicators, targeted attack | < 1 hour      | APT indicators, privilege escalation, data breach attempt | Shift lead or team           |
| Medium (P3)   | Suspicious activity requiring investigation        | < 2 hours     | Unusual lateral movement, malware execution               | Senior Analyst review        |
| Low (P4)      | Informational, potential false positive            | < 24 hours    | Policy violation, benign scan activity                    | Queue for batch review       |
| Info (P5)     | Contextual information, no action needed           | None          | Threat feed match with no local context                   | Log only                     |

### 2.2 Triage Decision Tree

Follow this decision tree for each incoming alert to determine appropriate disposition. First, assess whether the alert indicates an ongoing attack (critical/high) or a past/potential event (medium/low). Ongoing attacks trigger incident response protocol immediately—do not continue triage, escalate now. For non-ongoing events, evaluate the confidence score assigned by the detection engine (available in alert metadata). Score > 80%: investigate as true positive. Score 50-80%: quick enrichment (check threat intel, user context) then decide. Score < 50%: likely false positive, verify with 2-3 data points before closing. Always document triage reasoning in the alert comments—even "obvious" FPs need documentation for metrics and pattern analysis.

### 2.3 Enrichment Steps

Before making a final triage decision, enrich the alert with contextual data from integrated tools. Standard enrichment takes 2-5 minutes per alert and dramatically improves decision accuracy. Step 1: Check threat intelligence—query MISP and OpenCTI for IOCs (IPs, domains, hashes) from the alert. Match against known bad indicators = auto-confirm true positive. Step 2: Check user context—if alert involves a user account, verify identity, recent login locations, access patterns via LDAP/Active Directory integration. Anomalous user behavior = elevate severity. Step 3: Check asset context—is the affected asset critical (domain controller, payment server, database)? Critical asset = elevate severity regardless of alert score. Step 4: Check historical patterns—has this same alert fired for this same entity before? Repeated FP = close quickly. New pattern = investigate thoroughly.

### 3. Incident Response Workflow

When triage determines an alert (or combination of alerts) represents a genuine security incident requiring coordinated response, activate the incident response workflow defined in this section. The Djezzy SOC Platform follows NIST-inspired incident response lifecycle: Preparation, Detection & Analysis, Containment, Eradication, Recovery, and Lessons Learned. Each phase has specific activities, deliverables, and criteria for advancement to the next phase. The platform's SOAR integration (TheHive + Cortex) automates much of the workflow mechanics, but human judgment remains essential for complex decisions throughout the response process.

#### 3.1 Incident Creation in TheHive

Create a new case in TheHive to track the incident from detection through closure. From the alert detail view, click "Create Case" to pre-populate case fields with alert context (IOCs, observable artifacts, initial severity). Assign meaningful title format: "[Severity] [AttackType] BriefDescription [Date]" example: "[P2] Ransomware Detection FinanceServer 2026-07-30". Select appropriate case template (Malware, Data Breach, Phishing, DDoS, Insider Threat, Telecom Fraud, Other). Assign case owner (analyst who will lead investigation) and add tags for filtering (#critical, #apt, #ransomware, #ss7-fraud, etc.). Set task deadline based on severity SLA (P1: 4 hours, P2: 24 hours, P3: 72 hours, P4: 7 days). Click Create to initialize the case.

#### 3.2 Analysis Phase Activities

During analysis, gather sufficient information to understand what happened, how it happened, and what was affected. Document findings in case observations. Key activities: Timeline reconstruction—correlate timestamps across logs to build attack timeline (initial access → lateral movement → objective achievement). Scope determination—identify all affected systems, users, and data using GRR for endpoint forensics and Zeek/Arkime for network analysis. Attacker TTPs mapping—correlate observed behaviors to MITRE ATT&CK; framework techniques to understand attacker sophistication and predict next moves. Evidence preservation—create forensic images of affected endpoints before remediation (GRR flow: CreateImage). IOCs extraction—pull all indicators (IPs, domains, hashes, URLs) from the case for sharing via MISP and hunting across environment. Analysis phase ends when scope is understood and containment plan is ready.

#### 3.3 Containment Strategies

| Incident Type      | Containment Approach                     | Speed           | Business Impact                 | Example Use Case                  |
|--------------------|------------------------------------------|-----------------|---------------------------------|-----------------------------------|
| Malware infection  | Isolate affected host from network       | Minutes         | Host unavailable                | Ransomware on workstation         |
| Account compromise | Disable account, force password reset    | Seconds-Minutes | User locked out                 | Phished admin account             |
| Data exfiltration  | Block outbound C2 IPs, throttle egress   | Minutes         | Potential legit traffic blocked | Cloud storage upload detected     |
| DDoS attack        | Enable scrubbing, redirect DNS           | Minutes-Hours   | Latency increase                | Volumetric attack on web property |
| Insider threat     | Revoke access, preserve evidence         | Minutes         | Employee cannot work            | Data theft by departing employee  |
| Telecom fraud      | Block IMSI/MSISDN, disable fraud service | Seconds         | Legitimate subscriber affected  | SS7/SMSwap fraud in progress      |

## 4. Threat Hunting Operations

Threat hunting is the proactive search for threats that have evaded existing detections, complementing the reactive alert-driven workflow. Unlike incident response which responds to known-bad indicators, threat hunting forms hypotheses about adversary presence or behavior and tests them against available data. The Djezzy SOC Platform provides dedicated threat hunting workspace with hypothesis management, query building, and result tracking capabilities integrated with SIEM (Wazuh/Elasticsearch), EDR (GRR/Osquery), and threat intelligence (MISP/OpenCTI) data sources.

### 4.1 Hypothesis Development Framework

Effective threat hunts begin with well-formed hypotheses based on threat intelligence, industry trends, or internal observations. Use the hypothesis framework: "I believe [adversary/technique] may be present in our environment based on [reasoning], and I can detect it by looking for [data source/indicator]." Example hypotheses relevant to Djezzy's telecom environment: "I believe SS7 signaling abuse may be occurring based on recent industry reports of Diameter attacks against North African mobile operators, and I can detect it by looking for unusual MAP/ diameter message patterns in probe data." Another example: "I believe insider data theft may be occurring based on upcoming product launch with valuable IP, and I can detect it by looking for large data transfers to personal cloud storage from engineering workstations." Prioritize hypotheses by likelihood (based on intel) and impact (business criticality of targeted assets).

### 4.2 Hunting Using the Platform Workspace

Access the Threat Hunting workspace from the main navigation menu. Create a new hunt session with descriptive name reflecting the hypothesis (e.g., "SS7\_Diameter\_Abuse\_Hunt\_20260730"). Define the hunt scope: which data sources to query (SIEM events, EDR telemetry, network captures, telecom probes), time range (recommend starting with 30 days for initial hunt), and entities to focus on (specific subnets, user groups, device types). Write queries using the platform's unified query language (supports Lucene for Elasticsearch, SQL-like syntax for relational data, and YARA-L for behavioral rules). Execute queries iteratively—start broad, narrow based on results. Save interesting findings as observables linked to the hunt session. If hunt confirms hypothesis (finds true positives), convert to incident for formal response. If hunt disproves hypothesis (clean results), document negative finding—it's still valuable for detection gap analysis.

## 5. Threat Intelligence Management

Threat intelligence transforms raw data about threats into actionable knowledge that improves detection accuracy, speeds up incident response, and informs strategic security decisions. The Djezzy SOC Platform integrates two complementary threat intelligence platforms: MISP for tactical/operational IOCs (IPs, domains, hashes, URLs) and OpenCTI for strategic understanding (threat actors, campaigns, TTPs, kill chains). Together they provide a comprehensive view of the threat landscape relevant to Algerian telecommunications sector operations.

### 5.1 MISP Indicator Management

MISP serves as the platform's primary IOC repository, storing millions of indicators aggregated from commercial feeds, open source communities, and internal incident analysis. Daily operations include: reviewing newly published indicators from feeds (check Events > Automation > Feed status for failures), creating indicators from incident IOCs extracted during analysis (use Cortex analyzer for automatic extraction), searching indicators during alert triage (integrated lookup from alert detail view), and publishing internal indicators to trusted communities (telecom ISAC, national CERT). Maintain IOC quality by regularly expiring outdated indicators (TTL-based auto-expiration recommended), deduplicating overlapping entries, and updating sighting counts to reflect current relevance. Target IOC coverage: 95%+ of alerts should match at least one IOC for automated classification support.

### 5.2 OpenCTI Analysis Workflows

OpenCTI provides the analytical framework for understanding adversary behavior beyond individual IOCs. Use OpenCTI for: threat actor profiling—research APT groups targeting telecom sector (known telecom-focused APTs include APT41, Lazarus, Charming Kitten), campaign tracking—monitor ongoing campaigns affecting North African/MENA region, kill chain mapping—map observed TTPs to MITRE ATT&CK; to identify detection gaps, and diamond model analysis—construct victim-infrastructure-capability adversary models for complex investigations. Integrate OpenCTI with the platform's automated analysis pipeline: configure connectors to pull data from external sources (VirusTotal, Shodan, CISCO Talos), run Cortex analyzers on observables for enrichment, and export finished reports to TheHive cases for analyst consumption. Schedule weekly reviews of OpenCTI dashboards to identify emerging trends requiring detection rule updates.

## 6. Vulnerability Management Cycle

The vulnerability management program identifies, assesses, prioritizes, and remediates security weaknesses in the platform's infrastructure and the broader Djezzy IT estate. OpenVAS/GVM provides vulnerability scanning capability while DefectDojo tracks findings through remediation workflow. Weekly scans cover all internet-facing infrastructure and internal high-value assets (domain controllers, databases, payment systems). Monthly scans cover remaining internal infrastructure. Scan results auto-import to DefectDojo via API integration. Vulnerabilities are prioritized using CVSS score adjusted for exploitability (is there a known exploit?), asset criticality (what's the business impact if exploited?), and threat context (is this being actively attacked in our sector?). Critical/High vulnerabilities have 30-day remediation SLA, Medium has 90 days, Low has 180 days. Track compliance metrics: mean-time-to-remediate (MTTR), vulnerability density (vulns per asset), risk acceptance rate (vulns accepted vs fixed).

## 7. Compliance Reporting

The platform generates compliance reports aligned with ANSSI (French National Security Agency) frameworks and ARTP (Algerian Post and Telecommunications Regulatory Authority) requirements for telecommunications security operations. Monthly compliance reports cover: access control effectiveness (review of privileged access grants/revocations), change management compliance (were all changes approved, tested, documented?), incident response metrics (MTTD, MTTR by severity), vulnerability remediation progress (SLA compliance percentage), security awareness training completion rates, and third-party risk assessments (vendor security posture). Annual reports provide comprehensive evidence for regulatory audits. Use the platform's compliance dashboard (Compliance menu) to generate reports on demand. Export to PDF for distribution to stakeholders. Retain reports for minimum 7 years per Algerian data protection regulations.

## 8. AI/ML Operations

The AI automation subsystem enhances SOC capabilities through machine learning models performing anomaly detection, threat classification, predictive analytics, and automated response recommendations. AI Ops responsibilities include: monitoring model performance metrics (accuracy, precision, recall, F1-score) via the AI Dashboard, retraining models when performance degrades below threshold (accuracy < 90% triggers retraining workflow), reviewing automated response actions taken by the AI engine to ensure appropriateness (check AI Automation dashboard daily), managing model versions and rollback capability if new model performs worse than previous, and providing feedback loops—mark AI classifications as correct/incorrect to improve future predictions. The AI engine currently runs 12 production models covering: network anomaly detection, user behavior analytics, phishing URL classification, malware family prediction, fraud pattern recognition, and threat score prediction. Model inference latency should be < 100ms for real-time use cases. Monitor GPU utilization on analytics server—target < 80% average with headroom for burst processing.

## 9. Geomarketing Analytics



The geomarketing module provides location-aware security analytics leveraging Djezzy's cellular network infrastructure data. Key capabilities include: threat heatmaps showing geographic concentration of security events across Algeria's 58 wilayas (provinces), subscriber location tracking for fraud investigation (aggregate level, privacy-compliant), cell tower coverage analysis correlating security incidents with network characteristics, and geo-fenced alerting for zone-based notifications. Access geomarketing features from the main navigation under "Geomarketing Dashboard". View the Algeria map with color-coded threat density (green=low, yellow=moderate, orange=elevated, red=critical). Click on any wilaya to drill into detailed event list for that region. Use the subscriber tracker (requires additional authorization) to investigate fraud patterns showing unusual movement or clustering indicative of SIM box operations or roaming abuse. Geo data refreshes hourly from the probe data pipeline—real-time enough for operational use while protecting subscriber privacy through aggregation and anonymization.

## 10. Telecom Fraud Investigation

Telecommunications fraud represents a significant revenue loss vector and security concern specific to operators like Djezzy. The platform's specialized fraud detection module monitors for: SIM swap fraud (unauthorized SIM replacement to hijack accounts), International Revenue Share Fraud (IRSF) (premium-rate number pumping), Wangiri fraud (one-ring callback scams), PBX hacking (VoIP system compromise for toll fraud), CLI spoofing (caller ID manipulation for social engineering), and SS7/Diameter signaling abuse (location tracking, call interception). When fraud alert fires, investigate using the Telecom Dashboard which shows CDR analysis, call pattern visualization, and subscriber history. Correlate with geomarketing data to identify geographic patterns (fraud often clusters around border areas or specific cell towers). Coordinate with fraud prevention team (separate from SOC but closely aligned) for subscriber blocking and law enforcement referral if criminal activity confirmed. Document all fraud investigations thoroughly—evidence may be needed for legal proceedings or regulatory filings with ARTP.

## 11. Backup & Recovery Procedures

Robust backup strategy protects against data loss from hardware failure, human error, ransomware, or site-wide disasters. The platform implements a 3-2-1 backup strategy: 3 copies of data, on 2 different media types, with 1 copy offsite (DR site). Backup types: Full database dump nightly (PostgreSQL pg\_dump compressed), WAL archiving continuous (streamed to DR), application configuration snapshots (git-managed configs backed up to DR), Elasticsearch snapshots weekly (to object storage), container volume backups (daily incremental, weekly full). Test restore procedures quarterly: perform point-in-time recovery to random date within retention window, verify data integrity, measure recovery time against RTO objectives. Document all restore tests with results. Ransomware-specific protection: backups stored on immutable storage (WORM) for minimum 30 days, air-gapped tape copies monthly, offline cold storage of encryption keys (separate from online backups).

## 12. Escalation Matrix

| Scenario              | First Level            | Second Level          | Executive         | External              |
|-----------------------|------------------------|-----------------------|-------------------|-----------------------|
| P1 Critical Incident  | Shift Lead (immediate) | SOC Manager + IR Team | CISO (within 1hr) | Law Enforcement/Legal |
| Detection Gap Found   | Senior Analyst         | Detection Engineer    | SOC Manager       | Vendor Support        |
| Tool Outage           | On-call Engineer       | Platform Team Lead    | IT Director       | Vendor Support        |
| False Positive Spike  | Any Analyst            | Detection Engineer    | SOC Manager       | N/A                   |
| Intel Sharing Request | Intel Analyst          | SOC Manager           | Legal/Compliance  | ISAC/CERT             |
| Fraud Pattern Found   | Fraud Analyst          | Fraud Prevention Lead | Fraud Director    | ARTP/Police           |
| HR Issue (insider)    | SOC Manager            | HR + Legal            | CEO (if C-level)  | Law Enforcement       |
| Media Inquiry         | N/A                    | Communications        | CEO               | PR Agency/Legal       |

# Appendix A: Quick Reference Cards

## A.1 Useful Commands

```
Platform Status: curl -s https://soc.example.com/api/health | jq . Docker Services:
docker-compose -f docker-compose.prod.yml ps Container Logs: docker-compose -f
docker-compose.prod.yml logs -f [service-name] Database Query: psql -h db -U soc -d
soc_platform -c "SELECT * FROM incidents LIMIT 10;" Redis Check: redis-cli -a
[password] info server | head -20 Kafka Topics: kafka-topics.sh --bootstrap-server
kafka-0:9092 --list Grafana Dashboards: curl -u admin:admin
http://grafana:3000/api/search TheHive Cases: curl -u api-key:[key]
https://thehive:9000/api/case?range=0-10 MISP Events: curl -k
https://misp/events/index -H "Authorization: [key]" Wazuh Alerts: curl -k -u
admin:[pwd] https://wazuh:55000/alerts?limit=10
```

## A.2 Important File Locations

```
Platform Code: /opt/soc/platform/ Configuration: /opt/soc/config/ Environment:
/opt/soc/platform/.env.local Docker Compose:
/opt/soc/platform/docker-compose.prod.yml Application Logs: /opt/soc/logs/ PostgreSQL
Data: /var/lib/postgresql/16/main/ Redis Data: /var/lib/redis/ Kafka Data:
/var/lib/kafka/data/ Elasticsearch Data: /var/lib/elasticsearch/ Certificates:
/opt/soc/certs/ Backups: /opt/soc/backups/ Monitoring Config: /opt/soc/monitoring/
```

# Appendix B: Contact Directory

| Role                     | Name (Example)       | Contact            | Availability     |
|--------------------------|----------------------|--------------------|------------------|
| SOC Manager              | Ahmed Benali         | +213 555 0101      | 07:00-19:00      |
| Shift Lead (Day)         | Fatima Zahra         | +213 555 0102      | 07:00-19:00      |
| Shift Lead (Night)       | Karim Hadj           | +213 555 0103      | 19:00-07:00      |
| Platform Engineer        | Mohamed Ali          | +213 555 0104      | On-call rotation |
| Network Engineer         | Samia Bouazza        | +213 555 0105      | 09:00-17:00      |
| Database Admin           | Omar Khelifati       | +213 555 0106      | On-call rotation |
| CISO Office              | Director Security    | +213 555 0100      | Emergency only   |
| IT Helpdesk              | Support Team         | +213 555 0099      | 24/7             |
| Vendor Support (Wazuh)   | Wazuh Support Portal | support@wazuh.com  | Business hours   |
| Vendor Support (Elastic) | Elastic Support      | support@elastic.co | 24/7 Enterprise  |